Bloodo

Blood is not always short. Coordination is.

A WhatsApp-first, consent-based platform that helps licensed blood centres reach the right eligible donor within hours, without wearing donors out.

Swasth & Samavesh Bharat | Healthcare Innovation | Track 1: Product / Technology

Seva First Innovation Challenge 2026 · Team Bloodo

Submission at a glance

Steps 1, 2, 3, 5, 11

Title

Bloodo: WhatsApp-first donor mobilisation for blood centres

Theme

Swasth & Samavesh Bharat (Healthy and Inclusive Bharat)

Sub-theme

Healthcare Innovation

Track

Track 1: Product / Technology Solution

Development stage

Prototype: clickable console, simulated WhatsApp replies

Category

Undergraduate and above

Blood may be available. Reaching the right donor at the right time is the challenge.

Four failures today

1. Donor availability is not systematically tracked.
2. Families carry the burden of emergency donor searches.
3. Blood centres depend on fragmented communication.
4. Rare and recurring transfusion needs are hard to coordinate.

What this leads to

- Delayed donor confirmation.
- Repeated, unnecessary donor contact.
- Poor stock visibility and avoidable expiry.
- Uncertainty for patients and families.

Why Bloodo: turn scattered donor searches into a connected, trackable and sustainable blood mobilisation network.

A daily shortfall, worst where blood cannot be stored

Step 6

14.6M
units of whole blood needed in India each year

~1M
units short each year, about 7% of demand

77.5%
of platelet demand met; platelets last only 5 days

6.5 lakh
units wasted yearly from cold-chain and coordination gaps

How often it happens

Every day, in every district. Negative groups are hard to find, thalassemia patients need blood every few weeks, and collections dip in summer when student donors are away.

Small centres feel it most

Around 38% of blood centres collect fewer than 2,000 units a year, with little capacity for outreach.

Sources: national demand and supply estimate (NCBI PMC8986005); Transfusion Today, ISBT, July 2025.

How donors are found today, and with Bloodo

Step 6

Today

1. Hospital asks the family to “bring a donor”
2. Request is forwarded across WhatsApp groups
3. Staff phone the same willing donors again and again
4. No record of who replied, who came, or whether it worked

With Bloodo

1. Centre posts the need once, in one console
2. Bloodo shortlists eligible, nearby, rested, opted-in donors
3. Donors get a small WhatsApp wave and reply in one tap
4. Every reply, arrival and donation is tracked live

Staff today cannot see who is eligible, who is reachable, or whether anyone is coming.

Who is affected

Step 8

Patients

Dengue and cancer patients needing platelets, rare-group and thalassemia patients, mothers with heavy bleeding after delivery. Delay is the harm.

Families

Pushed to find a donor themselves, often in a strange city, within hours.

Voluntary donors

Called repeatedly by many people at once, so the most willing donors burn out and stop answering.

Blood centre staff

Spend hours phoning lists and forwarding messages instead of collecting and testing blood.

Small and rural centres

About 38% collect fewer than 2,000 units a year and have no staff for outreach.

Seasonal-dip districts

Collections fall in summer, exams and festivals, when demand does not.

Eight field-level difficulties

Step 6

1

Unorganised donor availability

No unified view of eligibility, location or history.

2

Family-driven searches

Families call relatives and strangers mid-emergency.

3

Fragmented centre operations

Donor lists, stock, calls and requests sit in separate places.

4

Rare and matched donors

Rare phenotypes need more than ABO and Rh searches.

5

Short shelf life

Platelets last about 5 days; usable units sit idle in one place, missing in another.

6

Rural access gaps

Distance, transport and communication limit reach.

7

Repeat-transfusion uncertainty

Thalassemia patients need dependable, recurring access.

8

Disconnected records

Donor and stock data are not visible across centres.

Each difficulty is matched to a Bloodo feature on slide 10.

The hidden cost of the problem

Step 6

One shortage, four consequences.

Time lost

Hours spent searching and calling donors.

Donor fatigue

Repeated requests discourage willing donors.

Information gaps

No unified view of donors, requests and stock.

Patient uncertainty

Delayed confirmation for urgent and recurring needs.

THE ACTUAL CONSEQUENCE

A willing donor, available blood and a waiting patient may never get connected in time.

One screen for the blood centre, five capabilities

Step 7

Staff pick the blood group, component, units and deadline. Bloodo finds eligible donors; licensed professionals make every eligibility and compatibility decision.

Smart Donor Matching

Compatible, recovered, nearby, opted-in donors are shortlisted before anyone is messaged.

Faster donor identification

Built

WhatsApp waves

Small batches with Yes, Not now and Pause buttons. No app to install.

Less calling for staff

Simulated

Donor protection

Monthly cap, rest after “not now”, quiet hours and one request at a time.

Donors keep saying yes

Built

Care Circles

A standing donor group per recurring-transfusion patient, asked in turn, booked 7 days ahead.

Predictable donor planning

Built

Expiry Rescue

Flags units nearing expiry and offers them to nearby centres that need them.

Better stock utilisation

Simulated

Built: works in the prototype. Simulated: runs on sample data or simulated replies.

Every difficulty has a Bloodo answer

Step 7

Difficulty	Bloodo feature	Effectiveness	Status
Unmanaged donors	Smart Donor Matching	Faster donor identification	Built
Family burden	Simple request system	Less manual searching	Built
Repeated calls	WhatsApp waves	Lower staff workload	Simulated
Rare blood needs	Rare Donor Registry	Better specialised discovery	Planned
Short platelet life	Expiry Rescue	Better stock utilisation	Simulated
Rural access gaps	SMS/IVR, regional network	Wider reach	Planned
Recurring transfusions	Care Circles	Predictable donor planning	Built
Fragmented records	Console and Grid	Connected operations	Console built

The Rare Donor Registry and the cross-centre Grid are planned; the prototype shows early versions.

How it works, and what each step gives

Step 10

1 Centre posts the need

Group, component, units, deadline and location.

One tracked request

2 Bloodo picks donors

Compatible, recovered, nearby, rested and opted in.

No wasted asks

3 WhatsApp wave goes out

A small batch first, more only if needed.

Fewer donors disturbed

4 Donor replies in one tap

Yes gets the location and a checklist. Not now gets rest.

Live status for staff

5 Staff mark the donation

Arrived, donated or deferred, from the console.

A reliable record

6 Loop closes

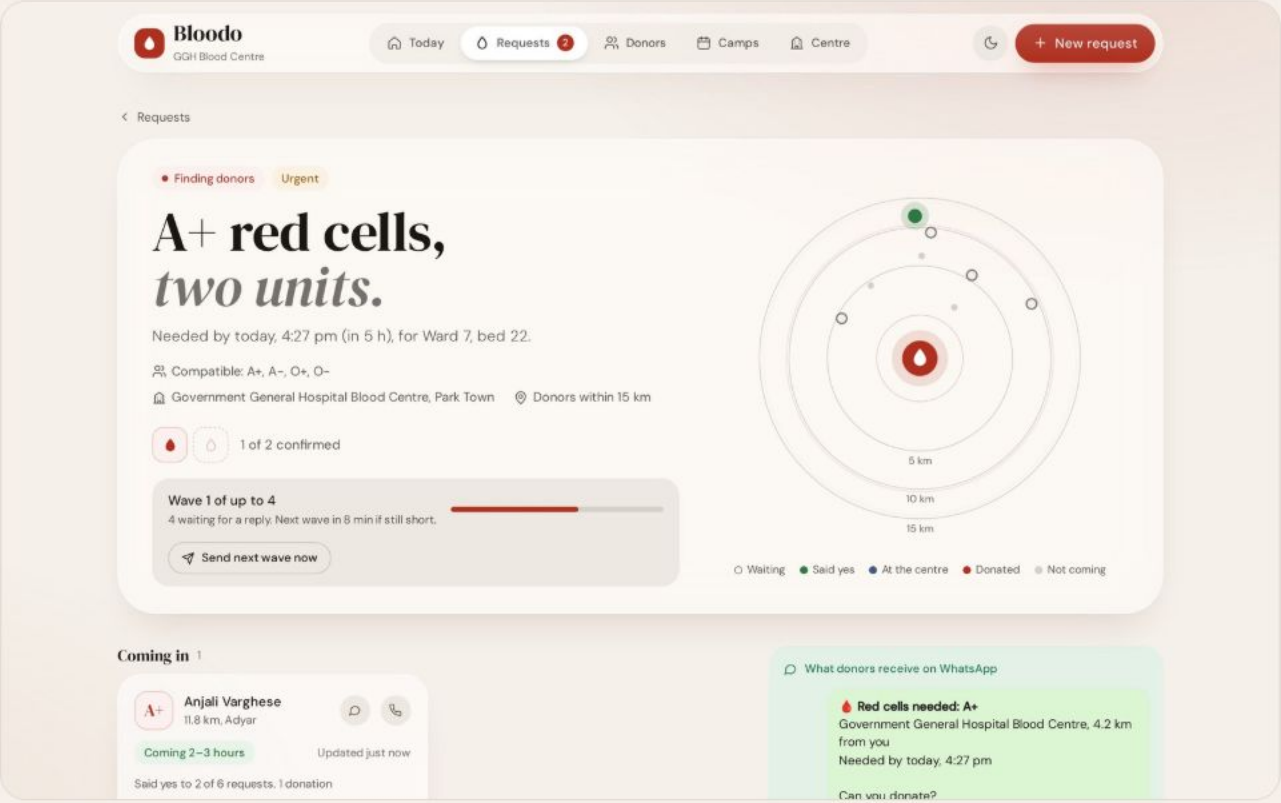
Thank-you, recovery timer and an “eligible again” reminder.

Donors come back

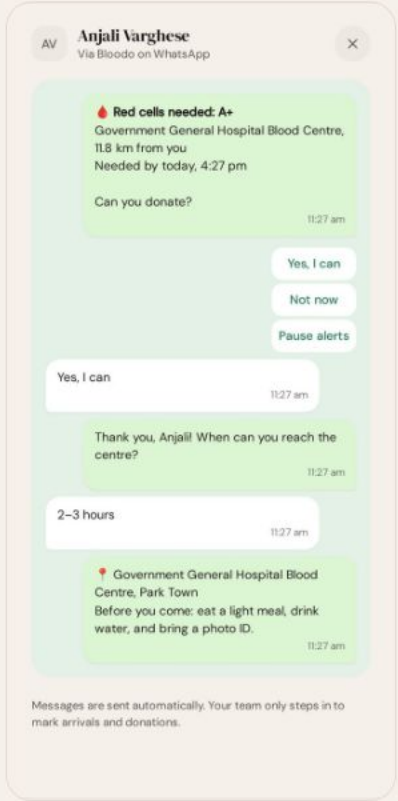
No reply in 20 minutes? The next wave goes out automatically. Covered? Everyone still waiting is told they are not needed, so no donor travels for nothing.

The prototype in action

Step 7



Live request board: donors sit on a distance map and change colour as they reply.



What the donor sees.

Working

Matching, waves, live board, Care Circles, stock, camps

Simulated

Donor replies, e-RaktKosh stock sync, other centres, masked calls

Screenshots are from the clickable prototype; no real patient or donor data is used.

What makes it different

	WhatsApp forwards	Donor directory apps	Bloodo
Knows who is eligible today	No	Partly	Yes
Targets donors in waves	No	No	Yes
Live replies and arrivals	No	No	Yes
Caps asks per donor	No	No	Yes
Needs no new donor app	Yes	No	Yes
Keeps donor numbers private	No	Partly	Yes
Verified, licensed centres	No	Partly	Yes

THE CORE IDEA

Treat a donor's goodwill as a limited resource.

Reach fewer, better-matched donors, and protect them so they keep saying yes.

Fair note: Bloodo complements e-RaktKosh and blood bank software for stock and records. It adds the outreach layer.

Comparison is our assessment of typical practice, to be validated in the field survey.

Donor protection, privacy and security

Step 9

2 requests a month, at most
Per donor, adjustable by the centre

14 days' rest after "not now"
Reasons update availability

Quiet hours, 10 pm to 7 am
Only critical requests at night

One request at a time
No donor asked twice at once

Control	What it does	Status
Consent	Explicit, purpose-specific, built to align with DPDP rules	Built (capture)
Privacy	Numbers never shown to others; patient details never in messages	Built
Masked calls	Staff and donors speak without seeing each other's numbers	Simulated
Access control	Role-based staff login and permissions	Planned
Encryption	In transit and at rest; WhatsApp Business Platform messaging	Planned
Audit trail	Who was asked, who replied, who donated, with timestamps	Activity log built

Licensed blood-centre professionals make every eligibility and compatibility decision.

Development stage: Prototype

Step 11

Idea (done) | **Prototype (now)** | Working prototype | Pilot | Deployed

Completed

- Clickable console: urgent requests, donors, camps, centre profile
- Matching by compatibility, recovery, distance and rest rules
- Waves and live reply tracking
- Care Circles with booking 7 days ahead
- Stock view with expiry offers and a 14-day outlook

Simulated today

- WhatsApp replies from donors
- Stock synced from e-RaktKosh
- Other centres accepting offers
- Masked calling

Still to do

- Connect the WhatsApp Business Platform
- Staff login, shared database, security hardening
- SMS fallback, IVR and bulk camp import
- Rare Donor Registry
- Field testing with a partner blood centre

An earlier donor app, Bloodo v1 (React and Firebase), shaped this pivot to blood centres.

Expected impact

Step 12

Health

Faster platelets, rare-group and matched units for patients who cannot wait, including thalassemia patients on regular transfusions.

Social

More voluntary, repeat donors and less pressure on families to find replacement donors.

Economic

Fewer staff hours on calls, less wastage, and messaging that costs about ₹1 per request.

Pilot targets we will measure

Within 6 hours

urgent requests filled

Measured: request sent to donor confirmed

2 or fewer

requests per donor each month

Measured: asks logged per donor

Every request

tracked from message to donation

Measured: share with a recorded outcome

These are targets, not results. No outcome has been proven yet; the pilot will test them.

Estimated cost for a one-district pilot

Step 13

Monthly running cost (3 centres, ~1,000 donors)	Per month
WhatsApp messages (~6,000 at ₹0.115)	₹700
WhatsApp provider platform fee	₹0 to ₹3,000
Cloud database and hosting	₹2,000 to ₹2,500
SMS fallback and masked calling	₹1,500
Monthly total	₹4,200 to ₹7,700

One-time costs	Amount
Development (student team)	In kind
Security and data protection review	₹30,000 to ₹50,000
Test devices and field visits	₹20,000

FIRST-YEAR PILOT TOTAL

₹1 to 1.6 lakh

12 months of running costs plus one-time costs.

About ₹1

of messaging per request: about 8 donors at ₹0.115 each. Replies inside 24 hours are free.

Estimates excl. GST. The ₹0.115 utility rate (Meta, India, 2026) is to be confirmed at launch.

From one district to many

Step 14

Phase 0

2–3 weeks

Secure login, shared database, WhatsApp Business verification

Phase 1

6–8 weeks

Pilot with 2–3 centres in one district, 300–500 donors from colleges and NSS

Phase 2

2–3 months

Eligibility reminders, SMS/IVR, platelet donor pools, Rare Donor Registry, standing groups for thalassemia patients

Phase 3

6–12 months

More districts, stock sharing between centres (Bloodo Grid), link with e-RaktKosh

Why it scales

No app to install

WhatsApp for most donors; SMS and IVR for the rest.

Cost grows with use

Pay per message, not per seat or per centre.

Local languages

Templates in Tamil, Malayalam, Hindi and more.

Interoperable

A coordination layer beside e-RaktKosh, not a replacement.

Evidence so far

Step 15 · Prototype, Other

Working prototype

A clickable console with the full request flow, simulated WhatsApp replies, donor and camp management, Care Circles and a stock view.

Published data

National demand, supply, platelet and wastage estimates from peer-reviewed and professional sources.

Earlier build

Bloodo v1, a donor registration and matching app on React and Firebase, which shaped this pivot to blood centres.

Next: field validation

A survey of blood centre staff and donors, then a 6–8 week pilot measuring time to fill and repeat donation.

What this shows: the workflow can be built and demonstrated. **What it does not yet show:** real donor response rates or time saved. The pilot is designed to measure both.

Who can adopt Bloodo, and what they gain

Step 16

State Blood Transfusion Councils

With the National Health Mission, to roll out across government centres.

Gain: network-wide visibility

District and medical college hospitals

The busiest centres, handling daily emergencies.

Gain: faster fills, less phoning

Indian Red Cross Society

Its blood centres and large voluntary donor networks.

Gain: donors reached without burnout

Private hospital blood centres

For surgeries, cancer care and platelet demand.

Gain: reliable platelet sourcing

Thalassemia and patient groups

To build standing donor groups for regular transfusions.

Gain: dependable recurring blood

NSS, NCC, Rotary and CSR teams

To run camps and register donors on the spot.

Gain: donors join ready to be asked

Support we need

Step 17

Support	What we need	What it unlocks
Mentorship	Transfusion medicine experts to review eligibility rules and workflows	Clinically sound rules
A pilot partner	One district or medical college blood centre for a 6–8 week pilot	Real-world evidence
Technical access	WhatsApp Business verification, cloud credits and a messaging provider	Live donor messaging
Legal guidance	A data protection review against the DPDP Act and Rules	Safe handling of donor data
Funding	About ₹1.6 lakh for the first-year pilot, including running costs and a security review	A funded, measured pilot

Bloodo

No patient should wait on a forwarded message.

We are ready to pilot with a blood centre in our district. We ask for a pilot partner, clinical mentors, WhatsApp Business access and about ₹1.6 lakh.

